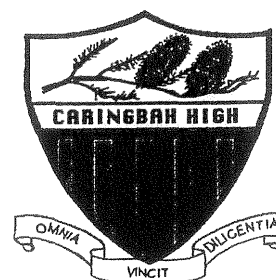


CARINGBAH HIGH SCHOOL

2011

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION



Mathematics Extension 2

General Instructions

Reading time - 5 minutes

Working time - 3 hours

Write using black or blue pen.

Board-approved calculators may be used.

A table of standard integrals is provided at the back of this paper.

Total marks - 120

Attempt Questions 1 - 8

All questions of equal value.

All necessary working should be shown in every question.

Question 1 (15 marks)**Marks**

(a) Find $\int \frac{dx}{(2x+1)^3}$ **2**

(b) Using integration by parts find the exact value of $\int_0^{\frac{1}{2}} \cos^{-1}x \, dx$. **3**

(c) Use the substitution $u=x-1$ to find $\int \frac{x}{\sqrt{x-1}} dx$ **3**

(d) Use the substitution $t = \tan \frac{x}{2}$ to find the exact value of **4**

$$\int_0^{\frac{\pi}{2}} \frac{1}{\sin x + 2} dx$$

(e) Evaluate $\int_0^1 \frac{5}{(2x+1)(2-x)} dx$ **3**

Question 2 (15 marks) Start a new page.

(a) Find the complex square roots of $7+6\sqrt{2}i$ giving your answer in the form $x+iy$ where x and y are real. **2**

(b) If $z = 3 + i$ find $\frac{i}{z}$ in the form $x + iy$. **2**

(c) Let $z_1 = 3 + 6i$ and $z_2 = -3 - 6i$.

Show that the locus specified by $|z - z_1| = 2|z - z_2|$ is a circle. **2**
Give its centre and radius.

(d) (i) Express $-1 + i$ in modulus-argument form. **1**

(ii) Express $(-1 + i)^6$ in the form $x + iy$. **2**

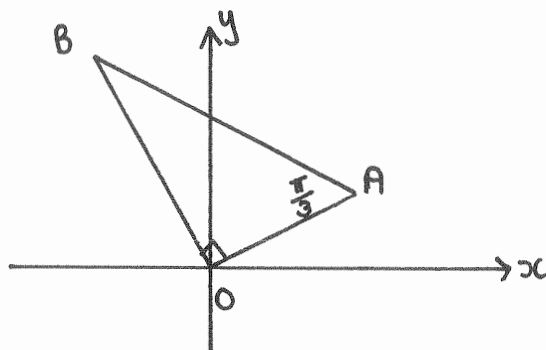
(e) Sketch the locus of z satisfying:

(i) $\arg(z + 2) = \frac{\pi}{4}$. **2**

(ii) $\operatorname{Re}(z) = |z|$. **2**

Question 2 continues on page 4

- (f) In the diagram below, the points A and B correspond to the complex numbers z and w respectively. $\angle AOB$ is a right angle and $\angle BAO = \frac{\pi}{3}$.

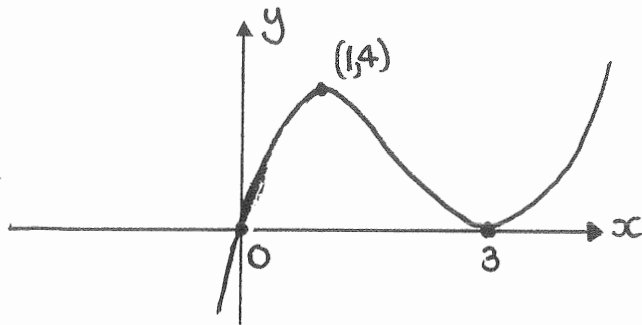


Show that $3z^2 + w^2 = 0$.

2

Question 3 (15 marks) Start a new page.

(a) The function defined by $g(x) = x(x-3)^2$ is drawn below.



Draw separate, one-third page sketches of :

(i) $y = g(|x|)$ 1

(ii) $y = \frac{1}{g(x)}$ 2

(iii) $y = \sqrt{g(x)}$ 2

(iv) $y = \tan^{-1}[g(x)]$ 2

(b) For the curve $x^2 + y^2 + xy - 4 = 0$:

(i) Find the x and y intercepts. 1

(ii) Using implicit differentiation show that $\frac{dy}{dx} = -\frac{2x+y}{x+2y}$ 2

(iii) Find any stationary points on the curve. 2

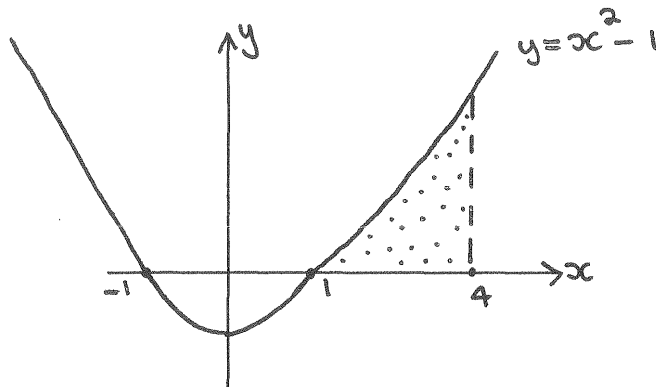
(iv) Deduce that the curve has vertical tangents at the points where $x = \pm \frac{4}{\sqrt{3}}$. 2

(v) Sketch the curve $x^2 + y^2 + xy - 4 = 0$. 1

Question 4 (15 marks) Start a new page.

(a)

4



The area bounded by the curve $y = x^2 - 1$, the x -axis and the line $x = 4$, as shown in the diagram, is rotated about the y -axis to form a solid. Use the method of cylindrical shells to find the volume of the solid.

(b) Sketch the graph of the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ showing the intercepts

4

on the axes, the coordinates of the foci and the equations of the directrices.

(c) The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with $a > b > 0$ has eccentricity e .

(i) Show that the line through the focus $S(ae, 0)$ which is

1

perpendicular to the asymptote $y = \frac{bx}{a}$ has equation

$$ax + by - a^2e = 0.$$

(ii) Show that this line meets the asymptote at a point on the corresponding directrix.

3

(d) Consider the polynomial $P(x) = x^3 - x^2 + x + 39$.

(i) Find the rational zero of $P(x)$.

1

(ii) Find the complex zeros of $P(x)$.

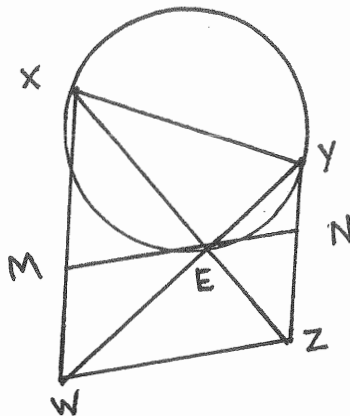
2

Question 5 (15 marks) Start a new page.

- (a) In the diagram below, $XYZW$ is a cyclic quadrilateral whose diagonals intersect at E . A circle is drawn through X , Y and E . MN is a tangent to this circle at E with M and N lying on XW and YZ respectively.

3

Copy this diagram.



Prove that MN is parallel to WZ .

- (b) Suppose that p and q are real numbers.

(i) Show that $pq \leq \frac{p^2 + q^2}{2}$.

2

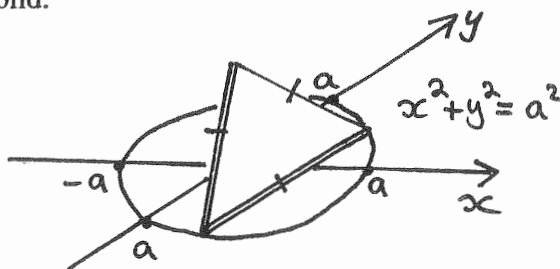
(ii) Hence show that for x and y real numbers $\frac{1}{xy} \leq \frac{x^2 + y^2}{2x^2y^2}$.

2

- (c) The base of a certain solid is the circle $x^2 + y^2 = a^2$.

3

Each cross-section of the solid is an equilateral triangle parallel to the y -axis with one side lying on the circle, as shown in the diagram. Find the volume of the solid.



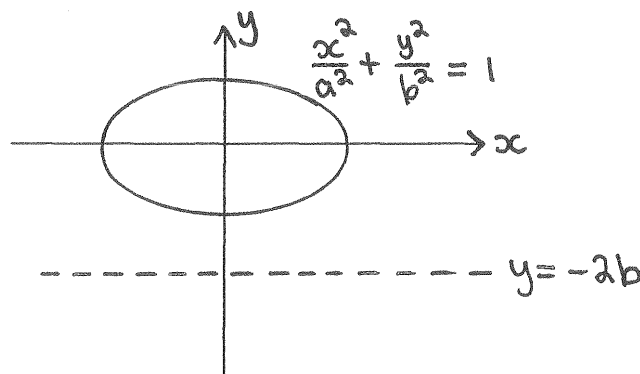
Question 5 continues on page 8

- (d) The complex cube roots of unity ω, ω^2 are two of the roots of **3**
 $P(x) = x^3 + px^2 + qx + r$.
Show that $p = q = r + 1$.

- (e) Resolve $\frac{1}{(x-3)(x^2+1)}$ into partial fractions over the **2**
field of real numbers.

Question 6 (15 marks) Start a new page.

- (a) The area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is rotated about the line $y = -2b$. A strip of thickness δx perpendicular to the axis of rotation sweeps out a slice whose cross-section is an annulus.



- (i) Show that this slice has a volume of $\delta V = 8\pi b y \delta x$. 2
- (ii) Hence find the volume of the solid which is formed. 3
- (b) (i) If $I_n = \int_{-1}^0 x^n (1+x)^{\frac{1}{2}} dx$ show that $I_n = -\frac{2n}{2n+3} I_{n-1}$. 3
- (ii) Hence evaluate I_3 . 2
- (c) By differentiating both sides of the formula: 3
- $$1 + x + x^2 + x^3 + \dots + x^n = \frac{x^{n+1} - 1}{x - 1} \quad \text{find an expression for:}$$
- $$1 + 2 \times 2 + 3 \times 4 + 4 \times 8 + \dots + n 2^{n-1}.$$
- (d) Given that $1 - 2i$ is a zero of the polynomial $p(x) = x^3 - 5x^2 + 11x - 15$ 2
factorise $p(x)$ over the field of complex numbers.

Question 7 (15 marks) Start a new page.

- (a) The normal at the point $P\left(cp, \frac{c}{p}\right)$ on the hyperbola $xy=c^2$ meets the x -axis at Q . M is the midpoint of PQ .
- (i) Show that the normal at P has the equation $p^3x - py = c(p^4 - 1)$. 2
- (ii) Show that M has coordinates $\left(\frac{c(2p^4 - 1)}{2p^3}, \frac{c}{2p}\right)$ 3
- (iii) Hence or otherwise, find the equation of the locus of M . 2
- (b) The numbers a , b and c are said to be in harmonic progression if their reciprocals $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in arithmetic progression, and b is then said to be the harmonic mean of a and c .
- (i) Show that the numbers 6, 8 and 12 are in harmonic progression. 1
- (ii) Show that the harmonic mean of a and c is $\frac{2ac}{a+c}$. 2
- (iii) If $a > 0, c > 0$ show that the geometric mean $\sqrt{ac}$ is greater than or equal to the harmonic mean $\frac{2ac}{a+c}$. 2
- (c) (i) Sketch $y = x^2 - 2x - 1$ showing the x -intercepts. 1
- (ii) Using mathematical induction and part (i) prove that $2^n > n^2$ for all integers $n \geq 5$. 2

Question 8 (15 marks) Start a new page.

- (a) The roots of the equation $x^3 + px + m = 0$ where $m \neq 0$ are α, β and δ . 2

Find an equation expressed in the form $ax^3 + bx^2 + cx + d = 0$ whose roots are α^{-2}, β^{-2} and δ^{-2} .

- (b) The vertices of a quadrilateral $ABCD$ lie on a circle radius r .
The angles subtended at the centre of the circle by sides AB, BC, CD and DA are respectively in an arithmetic progression with first term a and common difference d . (i.e. AB subtends an angle of a).

- (i) Show that $2a + 3d = \pi$ and interpret this result geometrically. 2

- (ii) Show that the area of the quadrilateral $ABCD$ is $2r^2 \cos d \cos \frac{d}{2}$. 3

[If required you may use the result: $\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}$]

- (c) Let $p = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$.

The complex number $\alpha = p + p^2 + p^4$ is a root of the quadratic equation $x^2 + ax + b = 0$, where a and b are real.

- (i) Prove that $1 + p + p^2 + \dots + p^6 = 0$. 2

- (ii) The second root of the quadratic equation is β . Justifying your answer, express β in terms of positive powers of p . 2

- (iii) Find the values of the coefficients a and b . 2

- (iv) Deduce that $-\sin \frac{\pi}{7} + \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{2}$. 2

End of paper

Ext.2 Trial HOC 2011

$$1. \textcircled{a} \int (2x+1)^{-3} dx = -\frac{1}{2} \cdot \frac{1}{2} (2x+1)^{-2} \\ = \frac{-1}{4(2x+1)^2} + C$$

$$\textcircled{b} u = \cos^{-1} x \quad \frac{dv}{dx} = 1 \quad \left[\begin{array}{l} \text{WE USE} \\ \text{INTEGRATION} \\ \text{BY PARTS} \end{array} \right. \\ \frac{du}{dx} = \frac{-1}{\sqrt{1-x^2}} \quad v = x$$

$$\int_0^{\frac{1}{2}} \cos^{-1} x \, dx = \left[x \cos^{-1} x \right]_0^{\frac{1}{2}} \\ - \int x \cdot \frac{-1}{\sqrt{1-x^2}} dx$$

$$= \left(\frac{1}{2} \cos^{-1} \frac{1}{2} - 0 \cdot \cos^{-1} 0 \right) + \int_0^{\frac{1}{2}} x (1-x^2)^{-\frac{1}{2}} dx \\ = \frac{1}{2} \times \frac{\pi}{3} - \left[(1-x^2)^{\frac{1}{2}} \right]_0^{\frac{1}{2}}$$

$$= \frac{\pi}{6} - \left(\frac{\sqrt{3}}{2} - 1 \right)$$

$$= 1 + \frac{\pi}{6} - \frac{\sqrt{3}}{2}$$

$$\textcircled{c} u = x-1 \Leftrightarrow x = u+1 \\ du = dx$$

$$\int \frac{u+1}{\sqrt{u}} du = \int u^{\frac{1}{2}} + u^{-\frac{1}{2}} du$$

$$= \frac{2}{3} u^{\frac{3}{2}} + 2u^{\frac{1}{2}} + C$$

$$= \frac{2}{3} (x-1)^{\frac{3}{2}} + 2(x-1)^{\frac{1}{2}} + C$$

$$= \frac{2}{3} \sqrt{(x-1)^3} + 2\sqrt{x-1} + C$$

$$\textcircled{d} t = \tan \frac{x}{2} \quad x=0, t=0$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} \quad x = \frac{\pi}{2}, t=1$$

$$= \frac{1}{2} (1 + \tan^2 \frac{x}{2})$$

$$= \frac{1}{2} (1+t^2) \quad \therefore dx = \frac{2dt}{1+t^2}$$

$$\text{Now } \sin x = \frac{2t}{1+t^2} :$$

$$\int_0^{\frac{\pi}{2}} \frac{1}{\sin x + 2} dx$$

$$= \int_0^1 \frac{1}{\frac{2t}{1+t^2} + 2} \times \frac{2dt}{1+t^2}$$

$$= \int_0^1 \frac{2dt}{2t + 2(1+t^2)}$$

$$= \int_0^1 \frac{dt}{t^2 + t + 1}$$

$$= \int_0^1 \frac{dt}{t^2 + t + \frac{1}{4} + \frac{3}{4}}$$

$$= \int_0^1 \frac{dt}{(t+\frac{1}{2})^2 + \frac{3}{4}}$$

$$= \int_0^{\frac{3}{2}} \frac{du}{u^2 + \frac{3}{4}}$$

$$= \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{u}{\frac{\sqrt{3}}{2}} \right]_{\frac{1}{2}}^{\frac{3}{2}}$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{2u}{\sqrt{3}} \right]_{\frac{1}{2}}^{\frac{3}{2}}$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right]$$

$$= \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6}$$

$$= \frac{\pi}{3\sqrt{3}}$$

$$\textcircled{e} \int_0^1 \frac{5}{(2x+1)(2-x)} dx = \int_0^1 \frac{2}{2x+1} + \frac{1}{2-x} dx$$

$$= \left[\ln(2x+1) - \ln(2-x) \right]_0^1$$

$$= \ln 3 + \ln 2$$

$$= \ln 6$$

$$\left[\begin{array}{l} \text{using } u = t + \frac{1}{2} \\ t=0, u = \frac{1}{2} \\ t=1, u = \frac{3}{2} \\ du = dt \end{array} \right.$$

2. a) $(x+iy)^2 = 7 + 6\sqrt{2}i$

$$x^2 - y^2 = 7 \text{ and } 2xy = 6\sqrt{2}$$

$$xy = 3\sqrt{2}$$

$$x = \pm 3 \text{ and } y = \pm \sqrt{2}$$

$$\text{ie. } 3 + i\sqrt{2} \text{ and } -3 - i\sqrt{2}$$

b) $\frac{i}{3+i} \times \frac{3-i}{3-i} = \frac{3i - i^2}{9 - i^2}$

$$= \frac{1 + 3i}{10}$$

$$= \frac{1}{10} + \frac{3i}{10}$$

c) $|(x+iy) - (3+6i)| = 2|(x+iy) - (-3-6i)|$

$$|(x-3) + (y-6)i| = 2|(x+3) + (y+6)i|$$

$$\sqrt{(x-3)^2 + (y-6)^2} = 2\sqrt{(x+3)^2 + (y+6)^2}$$

$$(x-3)^2 + (y-6)^2 = 4[(x+3)^2 + (y+6)^2]$$

$$(x+5)^2 + (y+10)^2 = 80$$

which is a circle with centre $(-5, -10)$ and radius $\sqrt{80}$

d) i) $(-1, 1)$ $\theta = \frac{3\pi}{4} \therefore \arg z = \frac{3\pi}{4}$

$$r = \sqrt{2}$$

$$\therefore z = \sqrt{2} \text{ cis } \frac{3\pi}{4}$$

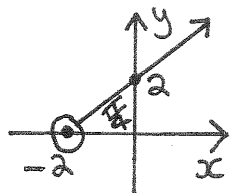
ii) $z^6 = (\sqrt{2})^6 \text{ cis } \frac{9\pi}{2}$

$$= 8 \text{ cis } \frac{\pi}{2}$$

$$= 8(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$$

$$= 0 + 8i$$

e) i)



ii) let $z = x + iy$

$$\text{Re}(z) = |z|$$

$$x = \sqrt{x^2 + y^2}$$

$$x^2 = x^2 + y^2$$

$$y = 0, x \geq 0$$



f) $w = \sqrt{3}iz$

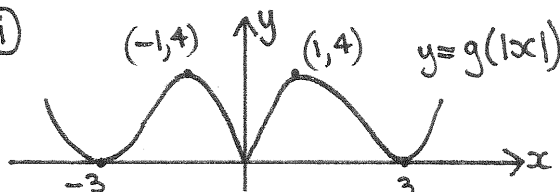
$$\therefore w^2 = 3i^2 z^2$$

$$w^2 = -3z^2$$

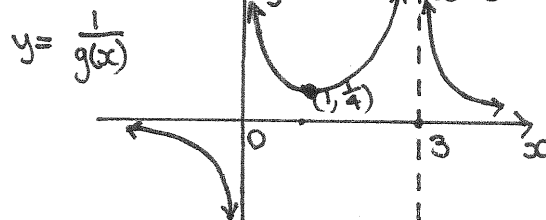
$$w^2 + 3z^2 = 0$$

$$[\tan \frac{\pi}{3} = \frac{OB}{OA} \Rightarrow OB = \sqrt{3}OA]$$

3 a) i)

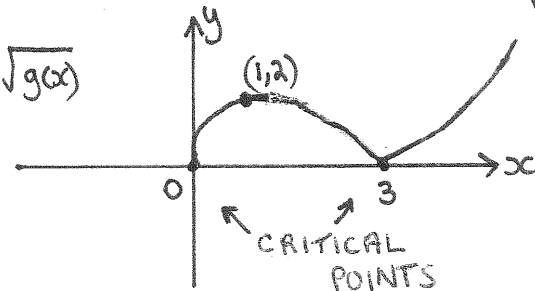


ii)

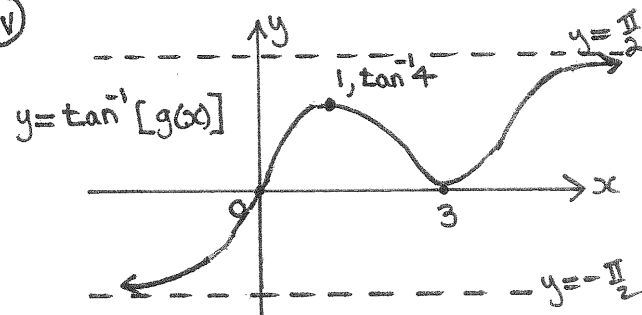


iii)

$$y = \sqrt{g(x)}$$



iv)



b) i) x-ints: ± 2 y-ints: ± 2

ii) $2x + 2y \frac{dy}{dx} + y \cdot 1 + x \frac{dy}{dx} = 0$

$$(x+2y) \frac{dy}{dx} = -(2x+y)$$

$$\frac{dy}{dx} = \frac{-(2x+y)}{x+2y}$$

iii) stat. points $\frac{dy}{dx} = 0 : 2x + y = 0$

$$y = -2x$$

sub. $y = -2x$ into eqn:

$$x^2 + 4x^2 - 2x^2 - 4 = 0$$

$$3x^2 = 4$$

$$x = \pm \frac{2}{\sqrt{3}}$$

$$\therefore \text{stat. points } (\frac{2}{\sqrt{3}}, -\frac{4}{\sqrt{3}}) \text{ and } (-\frac{2}{\sqrt{3}}, \frac{4}{\sqrt{3}})$$

(iv) vert. tangents: $\frac{dy}{dx}$ undefined

i.e. $x+2y=0$
 $y = -\frac{1}{2}x$

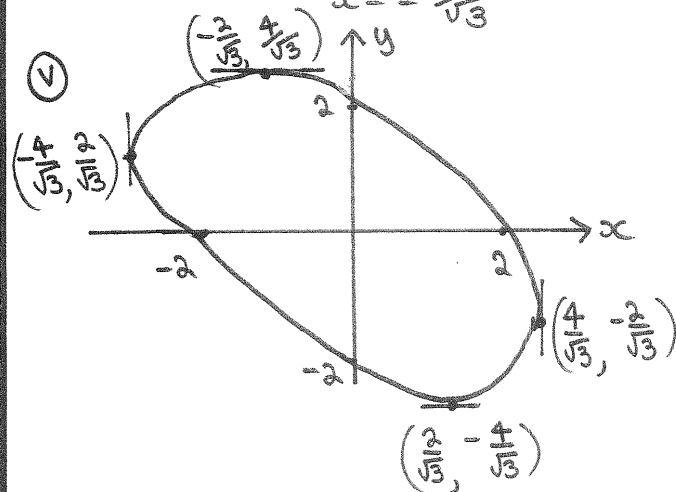
sub $y = -\frac{1}{2}x$ into eqn:

$$x^2 + \frac{1}{4}x^2 - \frac{1}{2}x^2 - 4 = 0$$

$$\frac{3}{4}x^2 = 4$$

$$x^2 = \frac{16}{3}$$

$$x = \pm \frac{4}{\sqrt{3}}$$



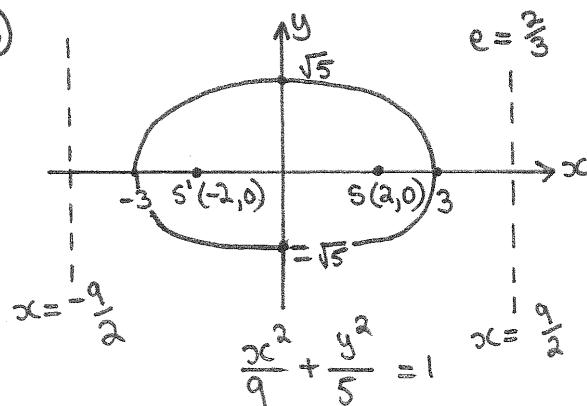
4. (a) $\delta V = 2\pi x (x^2 - 1) \delta x$

$$V = 2\pi \int_1^4 x^3 - x \, dx$$

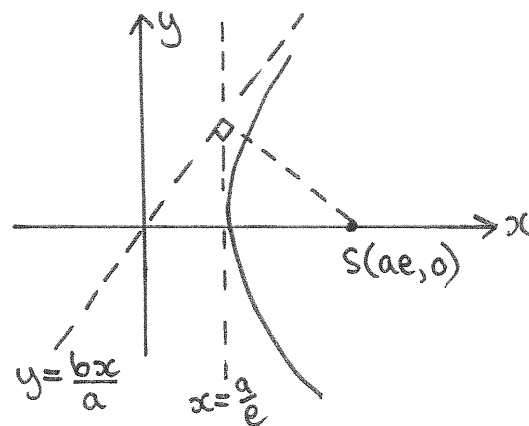
$$= 2\pi \left[\frac{1}{4}x^4 - \frac{1}{2}x^2 \right]_1^4$$

$$= \frac{225\pi}{2} \text{ units}^3$$

(b)



(c)



(i) gradient of line = $-\frac{a}{b}$

eqn of line: $y - 0 = -\frac{a}{b}(x - ae)$

$$ax + by - a^2e = 0$$

(ii) asymptote $y = \frac{b}{a}x$ and directrix

$x = \frac{a}{e}$ meet at $(\frac{a}{e}, \frac{b}{e})$

sub $(\frac{a}{e}, \frac{b}{e})$ into eqn of line:

$$\begin{aligned} \text{LHS} &= a \times \frac{a}{e} + b \times \frac{b}{e} - a^2e \\ &= \frac{a^2 + b^2}{e} - a^2e \end{aligned}$$

$$= \frac{a^2e^2}{e} - a^2e$$

$$= 0$$

$$= \text{RHS}$$

3

$$\begin{aligned} \text{SINCE } b^2 &= a^2(e^2 - 1) \\ b^2 &= a^2e^2 - a^2 \\ a^2 + b^2 &= a^2e^2 \end{aligned}$$

(d) (i) $P(x) = (x+3)(x^2 - 4x + 13)$

$\therefore$ ONLY RATIONAL ZERO IS $x = -3$

SINCE $x^2 - 4x + 13 = 0$ HAS NO RATIONAL ROOTS

(ii) NOW FOR $x^2 - 4x + 13 = 0$

$$x = \frac{4 \pm \sqrt{16 - 4 \cdot 1 \cdot 13}}{2}$$

$$= \frac{4 \pm \sqrt{-36}}{2}$$

$$= \frac{4 \pm \sqrt{36i^2}}{2}$$

$$= \frac{4 \pm 6i}{2}$$

$$= 2 + 3i \text{ and } 2 - 3i$$

$\therefore$ complex zeros: $-3, 2+3i, 2-3i$

5. (a) let $\angle EWZ = a^\circ$

$\therefore \angle ZXY = a^\circ$ (Ls in same segment circle $xyzw$)

$\therefore \angle YEN = a^\circ$ (alt. segment theorem)

$$\therefore \angle MEW = a^\circ \text{ (vert. opp. } \angle s)$$

$$\therefore MN \parallel WZ \text{ (altern. } \angle s, \angle MEW = \angle EWZ)$$

$$(b) \text{ (i) } (p-q)^2 \geq 0$$

$$p^2 - 2pq + q^2 \geq 0$$

$$p^2 + q^2 \geq 2pq$$

$$pq \leq \frac{p^2 + q^2}{2}$$

$$(ii) \text{ sub } p = \frac{1}{x}, q = \frac{1}{y} :$$

$$\frac{1}{x} \cdot \frac{1}{y} \geq \frac{\frac{1}{x^2} + \frac{1}{y^2}}{2}$$

$$\geq \frac{\frac{y^2}{x^2 y^2} + \frac{x^2}{x^2 y^2}}{2}$$

$$\geq \frac{x^2 + y^2}{2x^2 y^2}$$

$$(c) \text{ area } \Delta = \frac{1}{2} \cdot 2y \cdot 2y \cdot \sin \frac{\pi}{3}$$

$$= \sqrt{3} y^2$$

$$\text{vol. } \Delta = \sqrt{3} y^2 \delta x$$

$$\text{vol. solid} = \int_{-a}^a \sqrt{3} y^2 dx$$

$$= 2\sqrt{3} \int_0^a y^2 dx$$

$$= 2\sqrt{3} \int_0^a a^2 - x^2 dx$$

$$= 2\sqrt{3} \left[a^2 x - \frac{1}{3} x^3 \right]_0^a$$

$$= \frac{4\sqrt{3} a^3}{3} \text{ units}^3$$

$$(d) P(w) = w^3 + pw^2 + qw + r = 0$$

$$\text{i.e. } pw^2 + qw = -r - 1 \text{ since } w^3 = 1$$

$$P(w^2) = w^6 + pw^4 + qw^2 + r = 0$$

$$\text{i.e. } pw^4 + qw^2 = -r - 1 \text{ since } w^6 = (w^3)^2 = 1$$

$$\therefore pw^4 + qw^2 = pw^2 + qw$$

$$pw \cdot w^3 + qw^2 = pw^2 + qw$$

$$pw + qw^2 = pw^2 + qw$$

$$\therefore p = q$$

$$\text{If } p = q \text{ then } pw^2 + pw = -r - 1$$

$$p(w^2 + w) = -r - 1$$

$$-p = -r - 1 \text{ since } w^2 + w + 1 = 0$$

$$\text{i.e. } p = r + 1$$

$$\text{so } p = q = r + 1$$

$$(e) \frac{1}{(x-3)(x^2+1)} = \frac{a}{x-3} + \frac{bx+c}{x^2+1} \quad 4$$

$$1 = a(x^2+1) + (bx+c)(x-3)$$

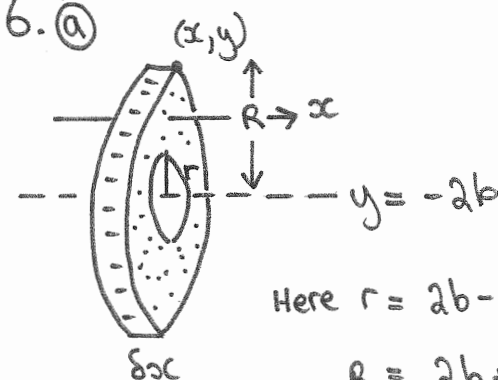
$$x=3: 1 = 10a \Rightarrow a = \frac{1}{10}$$

$$x=0: 1 = a - 3c \Rightarrow c = -\frac{3}{10}$$

$$\text{equate coeffs of } x^2: a+b=0 \Rightarrow b = -\frac{1}{10}$$

$$\text{i.e. } \frac{1}{10(x-3)} - \frac{x+3}{10(x^2+1)}$$

6. (a)



$$\text{Here } r = 2b - y$$

$$R = 2b + y$$

$$\text{annulus area} = \pi(R^2 - r^2)$$

$$= \pi[(2b+y)^2 - (2b-y)^2]$$

$$= \pi \times 8by$$

$$= 8\pi by$$

∴ volume of slice thickness δx is

$$\delta V = 8\pi b y \delta x$$

$$(ii) V = 8\pi b \int_{-a}^a y dx$$

$$= 8\pi b \int_{-a}^a \sqrt{\frac{b^2}{a^2}(a^2 - x^2)} dx$$

$$= \frac{8\pi b^2}{a} \int_{-a}^a \sqrt{a^2 - x^2} dx$$

$$= \frac{8\pi b^2}{a} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{a^2 - a^2 \sin^2 \theta} \cdot a \cos \theta d\theta$$

$$= \frac{8\pi b^2}{a} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} a \cos \theta \cdot a \cos \theta d\theta \quad \left\{ \begin{array}{l} \text{where } x = a \sin \theta \\ \text{so } dx = a \cos \theta d\theta \end{array} \right.$$

$$= \frac{8\pi b^2}{a} \cdot a^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$= 8\pi a b^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 2\theta) d\theta$$

$$= 4\pi a b^2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= 4\pi^2 a b^2 \text{ units}^3$$

$$(b) (i) u = x^n \quad \frac{dv}{dx} = (1+x)^{\frac{1}{2}}$$

$$\frac{du}{dx} = nx^{n-1} \quad v = \frac{2}{3} (1+x)^{\frac{3}{2}}$$

INTEGRATION BY PARTS:

$$I_n = \left[\frac{2}{3} x^n (1+x)^{\frac{3}{2}} \right]_{-1}^0$$

$$- \frac{2}{3} \int_{-1}^0 (1+x)^{\frac{3}{2}} n x^{n-1} dx$$

$$= 0 - \frac{2n}{3} \int_{-1}^0 x^{n-1} (1+x)^{\frac{1}{2}} (1+x) dx$$

$$= -\frac{2n}{3} \int_{-1}^0 x^{n-1} (1+x)^{\frac{1}{2}} + x^n (1+x)^{\frac{1}{2}} dx$$

$$= -\frac{2n}{3} \int_{-1}^0 x^{n-1} (1+x)^{\frac{1}{2}} dx - \frac{2n}{3} \int_{-1}^0 x^n (1+x)^{\frac{1}{2}} dx$$

$$= -\frac{2n}{3} I_{n-1} - \frac{2n}{3} I_n$$

$$\text{i.e. } \frac{2n}{3} I_n + I_n = -\frac{2n}{3} I_{n-1}$$

$$I_n \left(\frac{2n+3}{3} \right) = -\frac{2n}{3} I_{n-1}$$

$$I_n = \frac{-2n}{2n+3} I_{n-1}$$

$$(ii) I_3 = \frac{-6}{9} I_2$$

$$I_2 = \frac{-4}{7} I_1$$

$$I_1 = \frac{-2}{5} I_0$$

$$\text{Now } I_0 = \int_{-1}^0 x^0 (1+x)^{\frac{1}{2}} dx$$

$$= \frac{2}{3} [(1+x)^{\frac{3}{2}}]_{-1}^0$$

$$= \frac{2}{3}$$

$$\therefore I_3 = -\frac{6}{9} \times -\frac{4}{7} \times -\frac{2}{5} \times \frac{2}{3}$$

$$= -\frac{32}{315}$$

$$(c) 1 + 2x + 3x^2 + \dots + nx^{n-1}$$

$$= \frac{(x-1)(n+1)x^n - (x^{n+1} - 1) \cdot 1}{(x-1)^2}$$

[QUOTIENT RULE]

$$= \frac{(n+1)x^{n+1} - (n+1)x^n - x^{n+1} + 1}{(x-1)^2}$$

$$= \frac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}$$

let $x=2$:

$$1 + 2 \times 2 + 3 \times 4 + \dots + n 2^{n-1}$$

$$= \frac{n 2^{n+1} - (n+1) 2^n + 1}{(2-1)^2}$$

$$= n 2^{n+1} - (n+1) 2^n + 1$$

$$= 2^n (n-1) + 1$$

(d) other zero $1+2i$

$\therefore$ 2 factors are $[x - (1+2i)][x - (1-2i)]$

$$= x^2 - 2x + 5$$

and $x^2 - 2x + 5 \overline{) x^3 - 5x^2 + 11x - 15}$

i.e. $P(x) = (x-3)(x-(1+2i))(x-(1-2i))$

7. (a) (i) $y = c^2 x^{-1}$

$$y' = \frac{-c^2}{x^2}$$

At $(cp, \frac{c}{p})$: $m_{TAN} = \frac{-c^2}{c^2 p^2} = -\frac{1}{p^2}$

$\therefore m_{NORM} = p^2$

EQUATION NORM: $y - \frac{c}{p} = p^2(x - cp)$

$$p^3 x - py = c(p^4 - 1)$$

(ii) coords Q: sub $y=0$: $Q\left[\frac{c(p^4-1)}{p^3}, 0\right]$

x-coord of M = $\frac{cp^4 - p}{p^3} + cp$

$$\frac{2}{2}$$

$$= \frac{cp^4 - c}{p^3} + \frac{cp^4}{p^3}$$

$$\frac{2}{2}$$

$$= \frac{c(2p^4 - 1)}{2p^3}$$

y-coord of M = $0 + \frac{c}{p} = \frac{c}{2p}$

$\therefore M\left[\frac{c(2p^4-1)}{2p^3}, \frac{c}{2p}\right]$

(iii) sub $p = \frac{c}{2y}$ into $x = \frac{c(2p^4-1)}{2p^3}$

$$x = \frac{2 \times \frac{c^4}{16y^4} - c}{2 \times \frac{c^3}{8y^3}}$$

$$= \frac{\frac{c^5}{8y^4} - c}{\frac{c^3}{4y^3}} \times \frac{8y^4}{8y^4}$$

$$= \frac{c^5 - 8cy^4}{2c^3y}$$

$\therefore 2c^3xy + 8cy^4 = c^5$

$$2c^2xy + 8y^4 = c^4$$

(b) (i) i.e. show $\frac{1}{6}, \frac{1}{8}, \frac{1}{12}$ is an AP.

$$T_3 - T_2 = \frac{1}{12} - \frac{1}{8} = -\frac{1}{24}$$

$$T_2 - T_1 = \frac{1}{8} - \frac{1}{6} = -\frac{1}{24} = T_3 - T_2$$

$\therefore$ A.P.

$\therefore 6, 8, 12$ in harmonic progression

(ii) Find b if $\frac{1}{b} - \frac{1}{a} = \frac{1}{c} - \frac{1}{b}$

$$\frac{1}{b} + \frac{1}{b} = \frac{1}{a} + \frac{1}{c} \quad \frac{a}{ab} - \frac{b}{ab} = \frac{b}{bc} - \frac{c}{bc}$$

$$\frac{2}{b} = \frac{a+c}{ac}$$

$$b = \frac{2ac}{a+c}$$

$$\frac{a-b}{ab} = \frac{b-c}{bc}$$

$$\frac{a-b}{a} = \frac{b-c}{c}$$

$$ac - bc = ab - ac$$

$$2ac = b(a+c)$$

i.e. $b = \frac{2ac}{a+c}$

(iii) Show that $\sqrt{ac} - \frac{2ac}{a+c} \geq 0$

Now $\sqrt{ac} - \frac{2ac}{a+c} = \frac{\sqrt{ac}(a+c) - 2ac}{a+c}$

$$= \frac{\sqrt{ac}[a+c-2\sqrt{ac}]}{a+c}$$

$$= \frac{\sqrt{ac}[(\sqrt{a})^2 - 2\sqrt{a}\sqrt{c} + (\sqrt{c})^2]}{a+c}$$

$$= \frac{\sqrt{ac}[\sqrt{a} - \sqrt{c}]^2}{a+c}$$

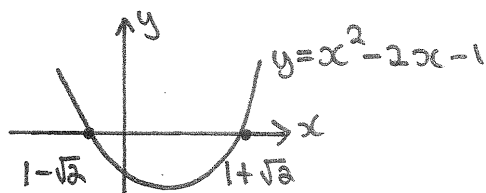
≥ 0 since $[\sqrt{a} - \sqrt{c}]^2 \geq 0$

and $a+c > 0$ for $a > 0, c > 0$

i.e. $\sqrt{ac} > \frac{2ac}{a+c}$

© (i) x -ints.: $x^2 - 2x - 1 = 0$

$x = 1 \pm \sqrt{2}$



(ii) STEP 1: Prove $S(5)$ true.

$2^5 = 32$

$5^2 = 25$

$\therefore 2^5 > 5^2$ i.e. $S(5)$ true.

STEP 2: Assume $S(k)$ true

i.e. $2^k > k^2$

Hence prove $S(k+1)$ true

i.e. $2^{k+1} > (k+1)^2$

or $2^{k+1} - (k+1)^2 > 0$.

Now $2^{k+1} - (k+1)^2 = 2 \cdot 2^k - (k+1)^2$
 $> 2 \cdot k^2 - (k+1)^2$
 by our assumption
 $> k^2 - 2k - 1$
 > 0 for $k > 1 + \sqrt{2}$

from part (i)

i.e. If $S(k)$ true then $S(k+1)$ true.

STEP 3: Hence if the result is true for $n=k$, it is true for $n=k+1$. It is true for $n=5$, so by the principle of mathematical induction it is true for all positive integers $n \geq 5$.

8. (a) eqn is $P\left(\frac{1}{\sqrt{x}}\right) = 0$

i.e. $\left(\frac{1}{\sqrt{x}}\right)^3 + \frac{p}{\sqrt{x}} + m = 0$

$\frac{1}{x\sqrt{x}} + \frac{p}{\sqrt{x}} + m = 0$

$x \times \sqrt{x}: 1 + px + mx\sqrt{x} = 0$

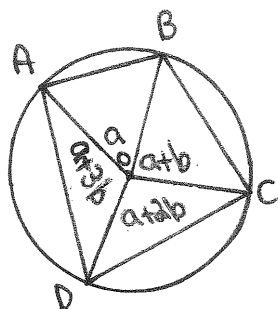
$1 + px = -mx\sqrt{x}$

$(1 + px)^2 = m^2 x^2 \cdot x$

$1 + 2px + p^2 x^2 = m^2 x^3$

$m^2 x^3 - p^2 x^2 - 2px - 1 = 0$

(b)



(i) $a + a + d + a + 2d + a + 3d = 2\pi$

($\angle$ s at a point)

$4a + 6d = 2\pi$

$2a + 3d = \pi$

$\therefore \angle BOD = 2a + 3d = \pi$

i.e. BD is a diameter.

(ii) $A = \frac{1}{2}r^2 \sin a + \frac{1}{2}r^2 \sin(a+d)$

$+ \frac{1}{2}r^2 \sin(a+2d) + \frac{1}{2}r^2 \sin(a+3d)$

$= \frac{1}{2}r^2 [\sin a + \sin(a+d) + \sin(\pi-a) + \sin(\pi-(a+d))]$

from $2a + 3d = \pi$

$a + 3d = \pi - a$

$a + 2d = \pi - (a+d)$

$= \frac{1}{2}r^2 [\sin a + \sin(a+d) + \sin a + \sin(a+d)]$

$= \frac{1}{2}r^2 [2\sin a + 2\sin(a+d)]$

$= r^2 [\sin a + \sin(a+d)]$

$= r^2 \left[2 \sin \frac{2a+d}{2} \cos \frac{d}{2} \right]$

$= 2r^2 \sin \frac{\pi - 2d}{2} \cos \frac{d}{2}$ from $2a+d = \pi - 2d$

$= 2r^2 \sin\left(\frac{\pi}{2} - d\right) \cos \frac{d}{2}$

$$= 2r^2 \cos d \cos \frac{d}{2}$$

$$\textcircled{C} \textcircled{i} \quad p^7 = \left(\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \right)^7$$

$$= \cos 2\pi + i \sin 2\pi \quad \text{by de Moivre}$$

$$= 1$$

$\therefore p$ is a root of $x^7 = 1$

$$\text{i.e. } x^7 - 1 = 0$$

$$(x-1)(1+x+x^2+\dots+x^6) = 0$$

Since $p \neq 1$ then $1+p+p^2+\dots+p^6 = 0$

$\textcircled{ii}$ Since coeffs of $x^2+ax+b=0$

then other root $\beta = \bar{\alpha}$

$$= \overline{p+p^2+p^4}$$

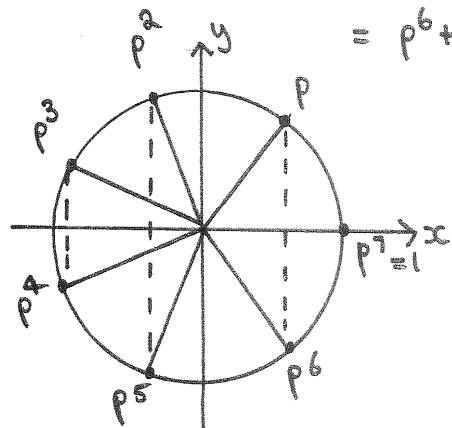
$$= \bar{p} + \bar{p}^2 + \bar{p}^4$$

$$= p^6 + p^5 + p^3$$

$$\text{since } \bar{p} = p^6$$

$$\bar{p}^2 = p^5$$

$$\bar{p}^4 = p^3$$



$$\textcircled{iii} \quad \alpha + \beta = -a$$

$$(p+p^2+p^4) + (p^6+p^5+p^3) = -a$$

$$p+p^2+\dots+p^6 = -a$$

$$-1 = -a \quad \text{since from } \textcircled{i}$$

$$1+p+p^2+\dots+p^6 = 0$$

$$\text{i.e. } a = 1$$

$$\alpha\beta = b$$

$$(p+p^2+p^4)(p^6+p^5+p^3) = b$$

$$p(1+p+p^3)p^3(p^3+p^2+1) = b$$

$$p^4(1+p+p^3)(1+p^2+p^3) = b$$

$$b = p^4(1+p^2+p^3+p+p^3+p^4+p^3+p^5+p^6)$$

$$= p^4(1+p+p^2+p^3+p^4+p^5+p^6+2p^3)$$

$$= p^4(0+2p^3) \quad \text{from } \textcircled{i}$$

$$= 2p^7$$

$$= 2 \times 1$$

$$= 2$$

$$\textcircled{iv} \quad \text{Now } p = \text{cis } \frac{2\pi}{7}$$

$$\therefore \alpha = p+p^2+p^4$$

$$= \text{cis } \frac{2\pi}{7} + \text{cis } \frac{4\pi}{7} + \text{cis } \frac{8\pi}{7}$$

$$\therefore \text{Im}(\alpha) = \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$$

$$= \sin \frac{2\pi}{7} + \sin(\pi - \frac{3\pi}{7}) + \sin(\pi + \frac{\pi}{7})$$

$$= \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7}$$

Also solving $x^2+ax+b=0$

$$\text{i.e. } x^2+x+2=0$$

$$x = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot 2}}{2}$$

$$= \frac{-1 \pm \sqrt{-7}}{2}$$

$$= \frac{-1 \pm \sqrt{7}i}{2}$$

$$= \frac{-1 \pm \sqrt{7}i}{2}$$

Taking $\alpha = \frac{-1 + \sqrt{7}i}{2}$ since from diagram $p+p^2+p^4 > 0$

$$\text{then } \text{Im}(\alpha) = \frac{\sqrt{7}}{2}$$

$$\text{i.e. } -\sin \frac{\pi}{7} + \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{2}$$